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Heterogeneous Effects of Generative artificial intelligence (GenAI) on Knowledge Seeking in Online Communities

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ABSTRACT

Generative AI (GenAI) may fundamentally reshape how users seek knowledge in online knowledge sharing communities. Although prior work found an overall decrease in knowledge seeking in online communities upon the availability of GenAI, the underlying dynamics across user groups have remained unexplored. This study addresses that gap. Drawing on commitment-based theory, we hypothesize that casual users—motivated by cost-benefit considerations—are more likely to reduce their question-posting activity than highly committed members. Using a *difference-in-differences* analysis, we find that ChatGPT's arrival leads to a substantial drop in questions on StackExchange, primarily driven by casual users (about 18.2%). Motivated by information foraging theory, we reveal heterogeneous downstream effects of GenAI on question characteristics. In particular, we find that the questions by casual users become more complex and novel, while those by intensive and top users do not. These results highlight the importance of heterogeneous user motivations in shaping platform dynamics, underscoring that while GenAI may diminish overall participation, it may also increase the value of the remaining content. Our study offers insights for knowledge sharing communities, managers, and stakeholders reliant on user-generated data, providing a nuanced view of GenAI's disruptive influence.

KEYWORDS

Generative AI; GAI; GenAI; knowledge seeking; commitment-based theory; information foraging theory; knowledge-sharing communities

Introduction

Digitization facilitates information exchange on a large scale, allowing members of society to connect at minimal cost. This has given rise to online knowledge sharing communities such as Quora, StackExchange, and Zhihu. Kuang et al. [37] define these communities as sites where individuals can simultaneously share their knowledge, experience, and expertise by posting and answering questions. Over the past two decades, such communities have enjoyed widespread adoption and proliferation. For example, with 418 million monthly visits, 3.1 million questions, and 3.5 million answers submitted, StackExchange (SE) is the World's largest knowledge sharing community [60]. Therefore, and not surprisingly, these

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online communities have become the focus of investigation by Information Systems, Management, and Economics scholars.

With the recent emergence of generative AI (GenAI) chatbots, the usage of knowledge sharing communities potentially awaits profound change. Typically, these GenAIs incorporate a large language model (LLM) into a chatbot capable of holding conversations and answering questions. As such, they enjoy increasing popularity among the knowledge seekers' community as answering devices, potentially reducing the usage of classical knowledge sharing community platforms. Early work in this direction has found that the introduction of GenAI, such as ChatGPT, reduces the number of questions posted to knowledge sharing communities [11, 19]. Yet, while these studies demonstrate an average effect of GenAI on knowledge seeking, there is still an insufficient understanding of whether GenAI has a heterogeneous effect across different user groups in such communities. This is somewhat surprising, given that there is a mature stream of literature on online communities, especially in information systems (IS), detailing diverse motivations to participate in community activities [5, 54, 46, 55, 72]. In particular, the commitment-based approach [5] contrasts continuance commitment—a type of need-based commitment wherein individuals weigh up benefits and costs—with affective and normative commitment—wherein users feel a strong emotional connection and loyalty to a community. Intuitively, this suggests that users of different commitment types may respond differently to the availability of GenAI. However, empirical evidence on this is still lacking, handicapping scholars, practitioners, and society alike.

While prior research [11, 19] has alerted us that GenAI stifles knowledge seeking in online communities, the implications of this contraction may need qualification based on the heterogeneous effects across user groups. In other words, the decrease in activity may not be as bleak as hitherto believed. Drawing on prior community literature [5, 46], heterogeneous effects are likely because users driven by cost-benefit calculus might be more susceptible to abandoning knowledge sharing communities in favor of GenAI than emotionally attached and loyal members. More importantly, information foraging theory [47] suggests that individuals engaging in cost-benefit calculus during their knowledge seeking will arguably need high benefit from asking questions that justify the effort of posting them to a community rather than using a low-effort GenAI. Plausibly, these high-benefit questions might be sophisticated ones that humans can answer more satisfactorily than GenAI. Against this backdrop, investigating the heterogeneous effects across user groups becomes a pressing need. Not only does it allow us to understand nuanced user group responses to GenAI, it also allows us to examine the granular downstream consequences for knowledge seeking on the platform.

This examination also matters for very practical reasons. Perhaps needless to say, the rise of GenAI may endanger the survivability of the platforms hosting these knowledge sharing communities. The threat to platforms may be more or less acute depending on whether highly committed and engaged users, as well as sophisticated content, are to be retained. Consequently, these platforms may continue to thrive as a valuable knowledge hub even after the availability of GenAI. Moreover, and dovetailing on this notion, the platforms might retain or enhance their value towards external stakeholders, such as companies looking to purchase training data for future GenAI versions.

Finally, investigating heterogeneous responses to GenAI in knowledge-sharing communities also has repercussions for society as a whole. If GenAI leads to a reshaping of

knowledge-sharing communities that leads to a growth in high-quality content, this may shift the power balance between the platform and the remaining users in favor of the users. To the extent that knowledge-sharing community members are also data workers [41, 40], the elevated importance of their high-quality contributions after the launch of GenAI may further fuel a debate on how much they should benefit from platforms monetizing their data.

Consequently, our study seeks to answer the following research question:

Research Question (RQ): How does the availability of Generative AI shape the knowledge seeking behavior of heterogeneous community groups in knowledge sharing communities?

To answer this question, we develop a set of twelve hypotheses drawing on the commitment-based approach [5]—an adaptation of commitment theory [44] to online communities—as well as on information foraging theory [47]. These theories are particularly suitable for our study because the commitment-based approach provides a theoretical angle on heterogeneous user motivations, and information foraging theory provides a theoretical framework to theorize on the downstream consequences for content characteristics provided to the platform after the introduction of GenAI.

We then test our hypotheses on rich data from the leading knowledge-sharing community StackExchange (SE) and leverage the launch of the ChatGPT GenAI as an exogenous shock. To identify a causal effect of ChatGPT on SE usage, we apply a novel difference-indifference (DiD) framework that leverages data from the year before the treatment as the control group and data from the treatment year as the treatment group [21]. Our findings are threefold and support our hypotheses. First, we find a more pronounced decrease in the volume of questions asked (approx. -18.2%) for casual users after the launch of ChatGPT and a softer decrease for intensive and top users. Second, we find that the complexity of questions, as measured by their readability, increases, which suggests that the questions posted to SE become more complex after ChatGPT. We show that casual users drive this increase, whereas the questions of intensive and top users do not increase in complexity. Finally, we find a concomitant rise in question novelty after the release of ChatGPT: questions after the availability of GenAI become more novel, driven by an increase in novel questions by casual users.

This paper proceeds as follows. In the next section, we review related literature and introduce our theoretical groundwork. Afterward, we develop our hypotheses. Then we introduce our dataset and estimation strategy before presenting the results of our hypotheses tests. Finally, we summarize our results and discuss their contribution to research and practice, outlining limitations and pointing out avenues for future research.

Related Literature

Effects of Generative AI

First, our paper relates to the emerging literature on the effects of GenAI, such as ChatGPT. Several studies investigate GenAI's potential to affect the economy overall. [22] find that 80% of the US workforce will have at least 10% of their tasks affected by GenAI. [25] have analyzed the jobs and industries most strongly exposed to GenAI. Their findings uncover a marked heterogeneity across jobs and industries, underlining

that the 20 most affected industries include software publishers, legal services, and travel agencies. Another set of studies investigates the potential of GenAI for textual data labeling. Gilardi et al. [27] find that ChatGPT is more effective in labeling text data than human agents. In their context, the zero-shot accuracy exceeds that of crowd workers for four of five labeling tasks. Scholars have also investigated how GenAI relates to content generation. 53. Reisenbichler et al. [53] show that semi-automated content generation search engine optimization using GPT-2 outperforms content generated by humans. ChatGPT-generated content may also affect how humans perceive content. ChatGPT, for example, proves effective for targeted ads, as ad copies automatically personalized by the GenAI tool are more persuasive than ads that were not personalized [58]. Another substream of literature underlines the potential danger of malicious purposes for GenAI use. In general, scholars have recognized that the democratization of content creation capabilities through GenAI comes with the drawback that the costs of production and proliferation have significantly decreased [12]. For example, GenAI can be used to compose propaganda texts that are as convincing as human-composed texts [28].

Finally, and perhaps most closely related to our study, research has started examining the effects of ChatGPT using data from SE [11, 19]. These studies have found an overall decrease in knowledge seeking, measured by the number of questions, after the availability of ChatGPT. Even though these studies have provided an important foundation, their results do not account for possible heterogeneous effects across user groups and resulting downstream changes in content characteristics. Our work extends the stream of research in this direction.

Online Communities and the Commitment-Based Approach

Although online communities are typically open to the general public and inclusive [24], past literature has recognized a certain degree of segmentation among online community participants [36, 39, 51]. 46. Oestreicher-Singer and Zalmanson [46] refer to this segmentation as the “ladder of participation,” wherein the participation of community members in community activities becomes more elaborate as their involvement with the community grows. In their four-tiered ladder, they recognize readers, contributors, collaborators, and leaders. Based on organizational commitment theory [44], Bateman et al. [5] propose an overarching theory to conceptualize online community commitment, which is termed the *commitment-based approach* (CBA). The CBA defines three types of commitment: continuance commitment, affective commitment, and normative commitment [5]. Each type is linked to distinct behavioral patterns and motives in online communities. Continuance commitment is defined as “an awareness of the costs associated with leaving an organization” [44:67]. As such, it is based on a cost-benefit calculus according to which members participate as long as the benefits of community participation outweigh its cost. However, they would cease participation if the same benefits were available elsewhere at lower costs. By contrast, affective commitment is defined as “emotional attachment to, identification with, and involvement in the organization” [44:67]. This commitment type is an emotional bond between a member and the community that is nurtured by shared goals and values [33], a sense of belonging [49], or trust-building social exchanges [16]. Finally, normative commitment is defined as “a feeling of obligation to continue employment” (p. 67) [44:67]

with an organization. It may arise from internalizing a sense of loyalty [67] or perceived indebtedness to the community that they are trying to reciprocate [38].

The CBA has been widely embraced by recent information systems (IS) literature on communities. For example, it was employed as a theoretical framework to explain that affective commitment can foster information sharing and seeking, which positively influences the engagement of online health community members [45]. Another stream of IS work has used the CBA to explain that social features on freemium websites can increase individuals' community commitment, which, in turn, affects the likelihood of paid subscriptions [46, 4]. A common denominator among all these studies is that they have viewed the CBA as a mechanism to grow and extend communities. We extend this view by shining a light on the CBA's potential as a mechanism of community resilience against GenAI competition. In particular, we will leverage it to develop hypotheses on the heterogeneous knowledge-seeking reactions driven by continuous, affective, and normative commitment to the availability of GenAI.

Information Foraging Theory

Information Foraging Theory (IFT) is an approach to analyzing human information search in information systems [47]. It is derived from Optimal Foraging Theory in biology and anthropology, which investigates the adaptive value of foraging strategies [61]. IFT examines the trade-off between the value of the information obtained and the costs of the activity in human-computer interaction tasks.

It posits that humans adapt their information seeking strategies to maximize information value and minimize costs, mirroring animals' optimization strategies in biological foraging [47]. When choosing which patch of land to roam, animals in the wild evaluate the trade-off between the effort it takes to achieve a certain amount of gain, such as caloric intake. Wolves decide which patch of land will yield the best prey-to-hunting time ratio, and deer decide which patch offers the best ratio between grass consumption and time spent. The theory emphasizes the importance of the external information environment, asserting that behavior is guided by rational principles that align with ecological constraints and the demands of the task [61]. In other words, it takes boundary constraints into consideration, for example, competitors.

IFT postulates that the human use of information systems to satisfy their information needs can be described similarly to the aforementioned animal foraging process. In deciding which

“patch of land,” i.e., which system (e.g., website, repository) to find the desired information, humans also weigh the value of the information gained with the effort, for example, time and cognitive effort, to obtain the information.

Past research in IS has repeatedly drawn on IFT to explain human information search processes. For example, it was found that sparse friendship networks provide non-redundant, valuable information on social networks. Thus, people will increase their exploitation effort to harness this valuable information, consistent with the IFT [30]. Moreover, IFT exhibited high external validity in predicting human web browsing behavior, as agent-based models designed on IFT principles closely resembled human web browsing patterns [42]. Subsequent studies also further illuminated that humans use cues or signals, referred to as “information scent,” such as hyperlinks, thumbnails,

or search results snippets, to identify and evaluate the information value of a resource [13, 48]. Extending this line of inquiry, we will draw on IFT to develop hypotheses on the usage decision of heterogeneous community members between GenAI and knowledge seeking communities.

Hypotheses Development

Following the literature on the “ladder of participation” [46]¹ and the “90-9-1”² rule [64], we start by categorizing community members into three groups, depending on their community activity. In ascending order of activity, we label these groups as casual, intensive, and top users. Next, we draw on the CBA [5] as well as on the conceptual mapping between the CBA and the ladder of participation, provided by Oestreicher-Singer and Zalmanson [46], to develop hypotheses on the responses to the availability of GenAI. We theorize that the idiosyncrasies of these groups’ behavioral patterns stipulated in the CBA will interact with the availability of GenAI. We note that the commitment types of the CBA are not mutually exclusive, and the boundaries are not clear-cut. Instead, it is likely that every user group has a different mix of commitment types. However, and consistent with the participation levels outlined in the ladder of participation [46], we build on the notion that casual users are particularly driven by continuous commitment, intensive users are mainly characterized by affective commitment, and top users are those with a high degree of normative commitment. Hereafter, for ease of reading, we will refer to continuance commitment-driven users as *casual users*, affective commitment-driven users as *intensive users*, and normative commitment-driven users as *top users*.

Drawing on the CBA, casual users interact with the community purely on a need-based motivation [55]. They follow a cost-benefit calculation when deciding whether to engage in community participation, and if the benefits outweigh the costs, they will participate. While other groups, such as intensive users, might value the social interaction itself, the casual users’ primary benefit of posting a question is to receive answers. Yet, posting a question is not without costs because questions must be carefully written following specific community standards. In addition, answers might take time to arrive, and heterogeneous answer quality might require additional effort to scan the usefulness of the answers [50]. Moreover, past literature points out that shyness represents another cognitive cost deterring casual users from posting and that they would rather browse existing content in the community before posting [50]. Hence, even in the absence of GenAI, these users will not post to the knowledge sharing community before exhausting other, lower-cost options such as searching older community posts, search engine queries, or real-life interpersonal exchange. The availability of GenAI impacts the calculus of casual users as it alters this cost-benefit computation. As per Meyer and Allen [44], these members will discontinue community commitment if the same benefits are available elsewhere at a lower cost. GenAI arguably decreases the cost of receiving answers, as phrasing questions does not have to follow community standards, and GenAI answers are more quickly available than community-sourced ones. Consistent with IFT [47], casual users’ knowledge seeking will be highly adaptive to the expected information value (i.e., answer quality) and the cost of accessing information (e.g., conforming to community guidelines, writing effort, waiting time). Consequently, they will have a relatively strong inclination to substitute community participation by GenAI.

In contrast, for intensive users, cost-benefit considerations play a subordinate role when deciding whether to maintain community participation. These users form an emotional attachment to the community through identification with the group [54, 5] and do not easily cut ties to a community due to easy-to-use outside options. Essentially, these users draw personal enjoyment from identifying and interacting with the community. Thus, their knowledge-seeking is not as strongly characterized by information foraging considerations (i.e., where can they get the highest-quality information at the lowest cost). However, while not cutting ties with the community entirely, GenAI also represents a valuable tool for these members, complementary to community participation. After all, while intensive users in this study might be emotionally attached to the community, to a smaller extent, they may be driven by continuance commitment that values the mere receiving of answers. Finally, we theorize that top users will be the least impacted by the availability of GenAI. To this group, using GenAI may be perceived as a norm violation or an act of disloyalty to the community. They are even more loyal to the platform than intensive users, for instance, due to stronger feelings of reciprocity towards the platform given their more intensive engagement with the community [55].³ The development of intense loyalty and reciprocity feelings is what propelled them to community activity levels higher than those of casual and intensive users. Therefore, we theorize that the knowledge seeking by normative commitment users will be even less characterized by information foraging principles, exhibiting the lowest tendency to substitute the community with GenAI. Consequently, we formulate our first set of hypotheses as follows:

Hypothesis 1 (HQ1): The availability of GenAI decreases the number of questions.⁴

Hypothesis 1a (HQ1a): The decrease in questions by casual users due to the availability of GenAI is stronger than the decrease by intensive and top users.

Hypothesis 1b (HQ1b): The decrease in questions by intensive users due to the availability of GenAI is weaker than the decrease for casual but stronger than the decrease for top users.

Hypothesis 1c (HQ1c): The decrease in questions by top users due to the availability of GenAI is weaker than the decrease for casual and intensive users.

Beyond the effects on the number of questions, a logical follow-up question pertains to the type of questions that community members will delegate to GenAI. For example, in terms of complexity, will they resort to GenAI for simple or complex questions? Per IFT [47], the need-based cost-benefit analysis that casual users conduct before posting to the knowledge-sharing platform imposes specific criteria on the question. Given that GenAI has decreased the cost of obtaining an answer of sufficient quality, it has thereby increased the relative cost of posting to the community. In plain terms, the benefit from receiving an answer of a given quality level will be higher using GenAI due to the lower information foraging costs than when posting to the community. While we conjecture that this will make it more unlikely for casual users to post to the community, it does not mean they will never engage with the community again. Conceivably, if the question is sufficiently complex, such that GenAI cannot yield an answer of satisfactory quality, the community remains a viable option.

Thus, based on the cost-benefit analysis of casual users postulated by the CBA [5], we theorize that while casual users will exhibit the strongest decrease in questions to the community (H1a), the questions they post to the community after the availability of GenAI will be more complex on average. If they are not complex enough, the cost-benefit considerations will unlikely yield higher benefits of posting the question to the community rather than using GenAI.

Turning to intensive users, the subordinate importance of elaborate cost-benefit analyses for these users before posting to the community does not impose strict complexity criteria on the questions when deciding to ask GenAI or the community. Even to the contrary, asking a question to GenAI might represent an opportunity cost in terms of enjoyment to these users. As they draw enjoyment from interacting and conversing with their peers in the knowledge sharing community [54, 5], they lose out on this enjoyment whenever they ask questions to GenAI instead of to the community. Thus, it supports the argument that intensive users will not systematically favor GenAI or the community for simple or complex questions. Nevertheless, although intensive users are primarily characterized by affective commitment, a slight tendency towards cost-benefit thinking might remain, making GenAI a good alternative for relatively simple questions. However, we expect that most of the questions would still be posted to the community. Therefore, we expect a moderate increase in question complexity by intensive users after the availability of GenAI, which is, however, weaker than the increase by casual users.

Finally, like intensive users, top users place low importance on cost-benefit analysis before contributing a question to the community. In their view, seeking knowledge in the community is, normatively speaking, the right thing to do, irrespective of the question's complexity. Therefore, one can argue that they will also not systematically select one or the other solution, GenAI or the community, for simple or complex questions. Yet, for very simple or even trivial questions, it is conceivable that top users will not have a strong feeling of norm violation when using GenAI. Analog to intensive users, this would sway them to post simple questions to GenAI, even though the majority of their questions would still be posted to the community. Consequently, we formulate our second set of hypotheses as follows:

Hypothesis 2 (HQ2): Question complexity increases due to the availability of GenAI.

Hypothesis 2a (HQ2a): The increase in question complexity by casual users due to the availability of GenAI is stronger than the increase by intensive and top users.

Hypothesis 2b (HQ2b): The increase in question complexity by intensive users due to the availability of GenAI is weaker than the increase for casual but not different from the increase by top users.

Hypothesis 2c (HQ2c): The increase in question complexity by top users due to the availability of GenAI is weaker than the increase for casual but not different from the increase by intensive users.

In past work, complex language has been recognized as a vehicle for knowledge generation and innovation. 6. Boland and Tenkasi [6] argue that in order to move from an

undifferentiated coarse problem statement to a more precise problem statement, complexification of language is required. Novel insights can be obtained by breaking an original problem into subproblems and developing a more accurate and coherent explanation of its subconstructs. In the past, this process has been vital to scientific discovery. Bradshaw [7] describes how the Wright brothers successfully used complexification to develop an airplane. While competitor teams framed the problem of manned flight using the metaphor of “chauffeurs of the air,” the wright brothers framed it as “flying a kite.” According to them, mastering manned flight required first understanding how and why a kite flew. Consequently, their kite framing allowed them to complexify the initial problem into subproblems; each focused on solving key challenges of manned flight, such as lateral control, sufficient lift, or reduced drag. As knowledge seekers engage in complexification, they critically depend on their language to support this cognitive process. That is, language needs to be increasingly complex to support the refinement of ideas and constructs. Beyond Boland and Tenkasi [6], other scholars have also recognized the role of language as a requirement for serving cognitive capacity and knowledge generation. For example, Bruner [8] postulates that what does not get structured narratively will be lost in memory. That is, to advance new knowledge, its subproblems have to be precisely defined, and this, in turn, requires more complex language. If the language does not support the process of complexification, it hampers the cognitive process necessary for the generation of novel knowledge. Consequently, based on the changes in complexity due to the availability of GenAI, we expect a concomitant increase in novelty.

Therefore, we formulate our final set of hypotheses as follows:

Hypothesis 3 (HQ3): Question novelty increases due to the availability of GenAI.

Hypothesis 3a (HQ3a): The increase in question novelty by casual users due to the availability of GenAI is stronger than the increase by intensive and top users.

Hypothesis 3b (HQ3b): The increase in question novelty by intensive users due to the availability of GenAI is weaker than the increase for casual but not different from the increase by top users.

Hypothesis 3c (HQ3c): The increase in question novelty by top users due to the availability of GenAI is weaker than the increase for casual but not different from the increase by intensive users.

Figure 1 provides an overview of our hypothesis in our research model.

Data and Method

Data

To test our hypotheses, we recover data from StackExchange (SE). SE is a network of many different knowledge sharing websites (hereafter *subforum*). Each subforum is dedicated to a different topic, and the most prominent subforum is StackOverflow (SO), mainly devoted

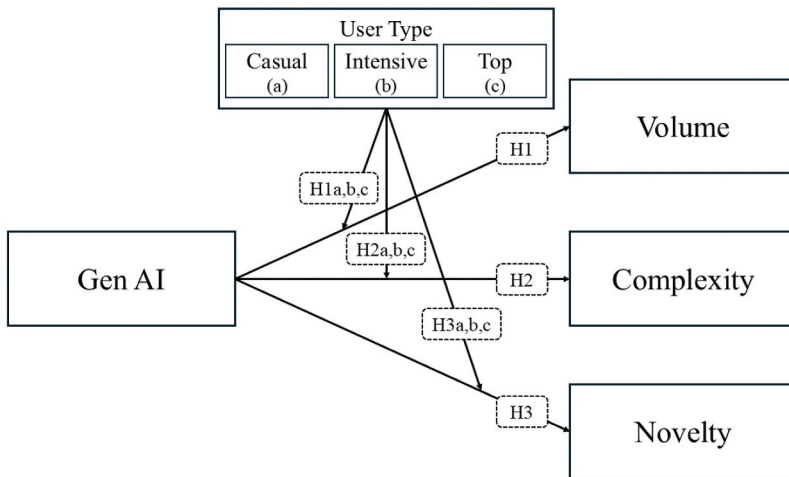


Figure 1. Research model.

to programming and software questions, Unix & Linux, and math. On the website, people can ask questions and receive answers from the community. With 418 million monthly visits, 3.1 million questions asked, and 3.5 million answers submitted, SE is the world's largest knowledge sharing platform. The website provides a Wikipedia-like editing feature for questions and a system of votes, badges, and user reputation management to ensure high-quality answers.

To construct our data set, we collect data from the top 50 SE subforums, including SO, between June 2021 and June 2023.⁵ We chose the large number of 50 subforums, as opposed to simply focusing on SO, to increase the generalizability of our dataset. We identify the top subforums based on their page views on the platform. We obtained this information from the publicly available SE Data Explorer.

Main Variables

Number of Questions

We collect all the questions for each subforum, including the title, the content of the question, the author ID, the primary tag, and the creation date. Our database contains 4,945,353 questions across 49⁶ subforums created between June 2021 and June 2023. Of those questions, 1,727,068 had already been moderated at the time of the data collection. In that case, SE allows us to recover the metadata of the question but not the text or the author ID. Therefore, we recover the question text of 3,218,285 questions. To test H1-1c, we aggregate the number of questions at the user level per week to gain insights into user heterogeneity.

User Type

To test H1-1c, we classify SE users into three types: casual, intensive, and top users. This approach is consistent with the tripartite community segmentation of our leading theory (the CBA) [5], as well as with the empirical observation that community engagement

follows Zipf's law, also known as the 90-9-1 rule [64, 14]. The 90-9-1 rule states that 90% of community members rarely post to a community, 9% of the members exhibit a medium amount of engagement, and 1% of the members provide the brunt of contributions. This further corroborates the ecological validity of the tripartite user group segmentation in addition to our focal theory. We classify treatment and control group users into one of the three categories, depending on how often they posted questions in the period between weeks 25 and 49, which is our pre-treatment period, that is, the time before the introduction of ChatGPT. This ensures that our classification is not affected by the introduction of ChatGPT. We define casual users as those who asked at least one question but fewer than three questions between weeks 25 and 49 in both the control and treatment periods. This captures users between the first and 85th percentiles regarding monthly questions asked and is thus broadly in line with Van Mierlo [64] and 14. Carron-Arthur et al. [14]. We define intensive users as those who posted between three and ten questions between weeks 25 and 49, comprising users between the 85th and the 99th percentiles in terms of monthly questions asked. Finally, we define top users as those users who posted more than ten questions between weeks 25 and 49, comprising the top 1% most active users in terms of questions asked.⁷

Question Complexity

To test H2-2c, we measure question complexity using readability indexes. This aligns with prior literature [23], which has used readability scores to measure the complexity of online review texts. On average, a complex question requires more cognitive skills to read and understand. Many readability indexes are available from past literature. To reduce the dependence on a single index, we compute a composite score by normalizing and averaging the scores of the following six different readability indexes: the Automated Readability (AR), the Flesch Reading Ease (FRE), the Flesch-Kincaid (FK), the Gunning-Fog (GF), the Simple Measure of Gobbledygook (SMOG), and the Coleman-Liau Index (CL). We measure this variable on the question level. Increasing scores in this variable indicate higher complexity.

Question Novelty

To test H3-3c, we measure question novelty by using the cosine similarity between question texts. The literature offers several techniques to compute the cosine similarity, such as term frequency-inverse document frequency (TF-IDF), latent semantic indexing (LSI), Doc2Vec [10], or Bidirectional Encoder Representations from Transformers (BERT) [35].

For the hypotheses tests, we employ a special BERT model, namely SentenceBERT (SBERT). This decision is based on recent literature [35] demonstrating that SBERT has the strongest correlation with human novelty assessments among contemporary approaches.⁸ SBERT operates as a fine-tuned transformer-based model designed explicitly for semantic similarity tasks. Unlike traditional embedding approaches, SBERT generates high-quality, dense vector representations of sentences, preserving their semantic and contextual meanings. We leverage the all-MiniLM-L6-v2 model, a lightweight variant of SBERT optimized for speed and performance.⁹ The model has been pre-trained on a large corpus of diverse texts, such as Wikipedia, enabling it to generalize well across domains. As prior research highlights significant thematic overlap between Wikipedia and StackExchange [65], using this model aligns well with our objective. As our measure of

novelty, we use the cosine similarity of the SBERT vector representations between two questions. We briefly outline this computation in the following section.

First, we define our reference set of questions by randomly sampling 100 questions after week 49 (i.e., the introduction week of ChatGPT) for each tag¹⁰ in both the control and treatment periods. Using the cosine similarity computed based on SBERT, we measure to what extent questions in the post-period (the reference set) are novel compared to questions in the pre-period with the same tag. Making the comparison within tags ensures that we do not obtain spurious novelty estimates. Were we to, for instance, compare the novelty across tags, we might get high novelty scores simply because we compare a *python* question with a question on how to cut a bonsai. Needless to say, the novelty would not be driven by the introduction of ChatGPT but, instead, would be the result of comparing apples with oranges. Cosine similarity is calculated for question pairs within the same tag. To form one question pair for each question in the reference set, we randomly sample 50 other questions with the same tag in the same group (treatment or control) asked before week 49, which gives us $100 \times 50 = 5,000$ unique pairs by tag, and $421 \times 100 \times 50 = 2,105,000$ observations across all tags. We then construct the novelty measure as the cosine similarity between the numerical vector representations of texts.

After and Treated

For our difference-in-difference analysis [2], we compute the following two variables. Following the difference-in-difference strategy by Eichenbaum et al. [21], *Treated* is a dummy variable that is 1 for the treatment group, that is, observations from between week 25 in 2022 to week 20 in 2023, and zero for the control group, that is, observations from week 25 in 2021 to week 20 in 2022. *After* is a dummy variable that is 1 after week 49 (the introduction of ChatGPT), spanning the time between week 49 and week 20. It is zero before the introduction of ChatGPT, that is, between weeks 25 and 49.

To obtain initial insights into the change in data quantity on SE and to illustrate our difference-in-difference framework before we formally introduce it in the next chapter, we plot the average number of questions aggregated at the tag level for the treatment (2022–23) and the control group (2021–22) in Figure 2. We can see that the number of questions for the control group is reasonably stable over the observation period, except for a marked decrease in activity during the week of Christmas.

We see a similar decrease for the period treatment group. Before the introduction of ChatGPT3, indicated by the dashed vertical line, we observed a similar trend in the evolution of questions for 2021–22 and 2022–23. After the introduction of ChatGPT3, there was a substantial decrease in questions in the treatment group.

General Identification Strategy

Scholars who study digital platforms are frequently confronted with shocks that affect all participants on a platform [66]. Identifying a causal effect using observational data in such settings is challenging. Past research has often relied on interrupted time series (ITS) analysis [29, 71]. ITS, however, requires two conditions. First, it requires a stable pre-treatment trend, but the stability of that trend is not testable in the absence of a control group. Second, platform-wide time-varying confounders must be ruled out.

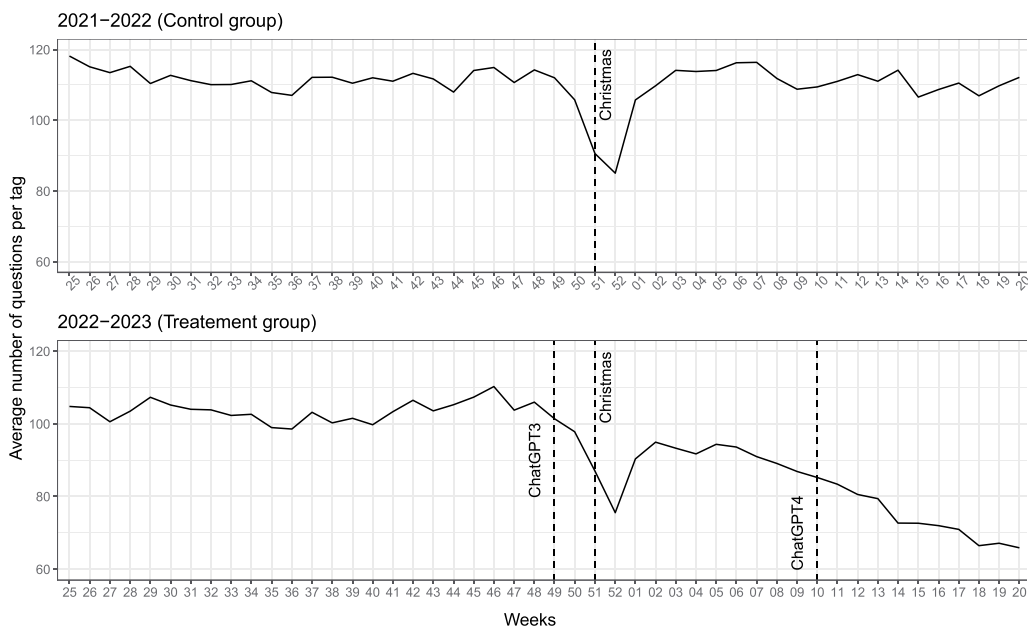


Figure 2. Model-free evidence of the effect of ChatGPT on the number of questions per tag.

To circumvent these limitations of ITS, we use a *difference-in-difference* style identification strategy, which employs temporal variation to create a control and a treatment group (as in [21, 26, 57]). In our case, and as ChatGPT launched on November 30, 2022, we assign data from June 2022 (week 25) to May 2023 (week 20) to be in the treatment group and data from June 2021 (week 25) to May 2022 (week 20) to be in the control group. This strategy contrasts the average change in community activity before and after the introduction of ChatGPT (November 30, 2022) between the treatment and the control group, which resembles the *difference-in-differences* strategy.

This strategy requires satisfying three assumptions to identify a causal effect of the introduction of ChatGPT [1, 21]. First, the 2021 and 2022 pre-treatment trends in the dependent variable before November 30 ought to be parallel, and this assumption is testable, unlike the stable pre-treatment trend in ITS. Figure 2 already suggests that this assumption is met, but we will formally test this assumption in the following section. Second, the launch of ChatGPT must be conditionally independent based on SE observables. This condition is satisfied as the introduction of ChatGPT has been exogenous and was not prompted by changes on SE. Third, the stable unit treatment value assumption must be satisfied, which requires the treatment to be qualitatively similar for all units and requires non-interference between the treatment and control groups. As 1) there are no variations in treatment, and 2) the temporal separation between the treatment and control groups rules out interference, this condition is satisfied.

Formally, this identification is denoted as follows in Eq. 1:

$$Y_* = \delta_0 \text{After}_w \times \text{Treated}_g + \delta_1 \text{Treated}_g + \delta_2 \text{AskWizard}_{wg} + \delta_3 \text{Copilot}_{wg} + \Psi + \varepsilon_{wg} \quad (1)$$

where Y_* represents the dependent variable (depending on which hypothesis is tested), and the coefficient δ_0 of the interaction term $\text{After}_w \times \text{Treated}_g$ captures the effect of ChatGPT on

the treatment group. The coefficient δ_1 of the variable $Treated_g$ estimates the mere difference between the treatment and control group before the introduction of ChatGPT.¹¹ One thing to note is that SE introduced a feature called “AskWizard” to its website shortly before the introduction of ChatGPT. This feature is there to help users, especially novices, to formulate questions and check whether the question they are posting has already been posted. To ensure that our results are not affected by the introduction of this tool [62], we control for *AskWizard*. Additionally, the software developer platform GitHub introduced “CoPilot,” an AI-based code completion tool, in October 2021. Plausibly, the launch of this tool might decrease the number of questions posted to SE. To ensure that this does not bias our estimation, we control for the launch of *CoPilot* using a dummy variable that is 1 for questions on programming languages supported by *CoPilot* (*python*, *javascript*, *typescript*, *ruby*) after the launch and 0 otherwise. Ψ captures a set of fixed effects depending on the analysis. It comprises week-of-the-year fixed effects controlling for SE activity in certain weeks of the year, for instance, during exam periods. When testing H1-1c, Ψ comprises user-level fixed effects to control for time-constant user-level heterogeneity because the unit of analysis is the user. When testing H2-2c and H3-3c, the unit analysis becomes at the question level. In this case, Ψ includes tag fixed effects that allow us to control for idiosyncratic differences between tags; that is, we can control for the fact that there are more questions related to *python* than to *java*. We additionally include tag-week fixed-effects, which control for time-varying tag-specific shocks, or in other words, sudden changes in activity in a given week of the year for a given topic. For example, a programming language might receive a surge in questions due to students’ exams in December or due to widespread software updates, such as iOS updates, in September. Finally, ϵ_{wg} is a random error term that follows the usual standard assumptions.

Results

Next, we present the estimation results that test our hypotheses.

Hypotheses 1-1c: Number of Questions for Casual versus Advanced versus Top Users

To test H1-1c, we balance the panel data by imputing the value of 0 for weeks during which no questions for a given user were asked. As we have 1,376,786 unique users in our dataset, this gives us $1,376,786 \times 96 = 132,171,456$ observations. However, to balance the panel even more and leverage user fixed effects, we only retain users who asked at least one question in the period before week 48 of their respective treatment group.¹² In the end, our database is composed of 46,699,056 observations. Given the count data nature of the dependent variable *Number of Questions* and the non-negligible number of zeros, we implement Equation 1 using a Poisson fixed effects model. Following prior literature [17, 9, 32, 3], this model lends itself naturally to the study of count data with zeros.¹³ Thus, the model presented in Equation 1 transforms into the following form:

$$E[Y_{uwg}|X_{uwg}] = \exp \left[\begin{aligned} &\delta_0 After_w \times Treated_g \times UserType_u + \delta_1 After_w \times Treated_g \\ &+ \delta_2 Treated_g \times UserType_u + \delta_3 After_w \times UserType_u + \delta_4 Treated_g \\ &+ \delta_5 After_w + \delta_6 UserType_u + \delta_7 AskWizard_{wg} + \mu_w + \kappa_u \end{aligned} \right] \quad (2)$$

where Y_{uwg} is the number of questions asked by user u in the week-of-the-year w in group g (control or treated). To test for differences across user types (H1a-1c), we use a triple interaction of . Owing to the user aggregation level, we add user-level fixed effects κ_u and week-of-the-year fixed effects μ_w . We present the results in Table 1.¹⁴ Column (1) presents the results for all users. Consistent with prior literature [11, 19], we find a statistically significant and negative overall effect, supporting H1. To test H1a-c, we need to test for the differences in the changes between the three different groups. This is done in Columns (2) to (4) using interactions. We change the reference group in each column to match the reference group in the respective hypotheses. For instance, in H1a, the reference group are casual users, hence they are the omitted reference group in Column (2).

The coefficient of $Treated \times After$ represents the effect on the reference group. The estimates in Column (2) suggest that casual users decrease their question asking-activity by $100(e^{-0.2009} - 1) = -18.2\%$. The coefficient of the interaction term $Treated \times After \times Intensive$ shows a statistically significant positive moderation for intensive users. Based on the estimates, intensive users significantly decrease their question-asking, but $100(e^{-0.2009+0.0600} - e^{-0.2009}) = 5.05\%$ ¹⁵ less than the casual users' initial decrease. We also find a positive moderation, although not significant on the 5% but only at the 10%-level, for top users, who decrease their number of questions asked by $100(e^{-0.2009+0.0419} - e^{-0.2009}) = 3.5\%$ less than casual users' decrease. Therefore, we find partial support for H1a because casual users decrease the number of questions asked statistically significantly more than intensive but not more than top users. From this, we deduce that GenAI has a stronger impact on community members driven by continuance commitment than on members driven by affective and normative commitment. Next, Column (3) tests H1b. The coefficient of $Treated \times After$ shows that intensive users decrease their question-asking activity by $100(e^{-0.1410} - 1) = -13.15\%$. When looking at the moderation for casual users $Treated \times After \times Casual$, we find again that casual users significantly decrease their question activity more than intensive users by performing the same computation (with intensive

Table 1. Effect of ChatGPT on the number of questions across user types on StackExchange.

DV	Number of Questions (user level)			
	(1)	(2)	(3)	(4)
Treated	-0.1468*** (0.0045)	-0.1145*** (0.0038)	-0.1375*** (0.0058)	-0.1606*** (0.0158)
Treated $\times$ After	-0.1784*** (0.0059)	-0.2009*** (0.0074)	-0.1410*** (0.0098)	-0.1590*** (0.0203)
Treated $\times$ After $\times$ Casual			-0.0600*** (0.0117)	-0.0419† (0.0216)
Treated $\times$ After $\times$ Intensive		0.0600*** (0.0117)		0.0181 (0.0229)
Treated $\times$ After $\times$ Top		0.0419† (0.0216)	-0.0181 (0.0229)	
User FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
AskWizard	Yes	Yes	Yes	Yes
User type FE	Yes	Yes	Yes	Yes
Ref. category		Casual	Intensive	Top
Hypothesis test	H1	H1a	H1b	H1c
Observations	46,699,056	46,699,056	46,699,056	46,699,056
Pseudo R ²	0.14753	0.1516	0.1516	0.1516

Standard errors at the user level in brackets; Q. before (x) by user type: Casual $0 < x < 3$; Intensive $2 < x < 11$; Top $> x$. †p < 0.1. ***p < 0.001. Abbreviation: FE, Fixed Effects; H1, Hypothesis 1.

users as the reference), namely by $100(e^{-0.1410-0.0600} - e^{-0.1410}) = -5.05\%$. However, the decrease in question for intensive users is not significantly different from top users, as shown by the statistically insignificant coefficient of $Treated \times After \times Top$ in Column (3). Therefore, we find partial support for H1b because although the decrease in questions for intensive users is not as strong as for casual users, it is as strong as the decrease for top users.

Finally, the estimates in Column (4) test H1c. The coefficient of $Treated \times After$ shows that top users decrease their question-asking activity by $100(e^{-0.1590} - 1) = -14.70\%$. We perform the same computation as for H1a but with top users as the reference and find that casual users decrease their question-asking activity more strongly by around $100(e^{-0.1590-0.0419} - e^{-0.1590}) = -3.5\%$, relative to top users, albeit not statistically significant at the 5%-level. Yet, we note that the difference between casual and top users is statistically significant at the 1%-level in the robustness checks that employ a negative binomial model (see Table 1.1 in Online Supplemental Appendix 1). Consequently, we find partial support for Hypothesis 1c.¹⁶ Again, the decrease in questions asking for top and intensive users is not statistically significantly different. Figure 3 summarizes our hypotheses tests visually.

Hypotheses 2-2c: Question Complexity

Next, we test H2-2c to examine whether the availability of ChatGPT also affected the complexity of questions on SE, as captured by the readability score. To this end, our depending variable becomes *Question Complexity* and Eq. 1 transforms into:

$$Y_{itwgu} = \delta_0 After_w \times Treated_g \times UserType_u + \delta_1 After_w \times Treated_g + \delta_2 Treated_g \times UserType_u + \delta_3 After_w \times UserType_u + \delta_4 Treated_g + \delta_5 After_w + \delta_6 UserType_u + \delta_7 AskWizard_{wg} + \delta_8 CoPilot_{wtg} + \mu_w + \sigma_t + \mu_w \times \sigma_t + \epsilon_{itwgu} \quad (3)$$

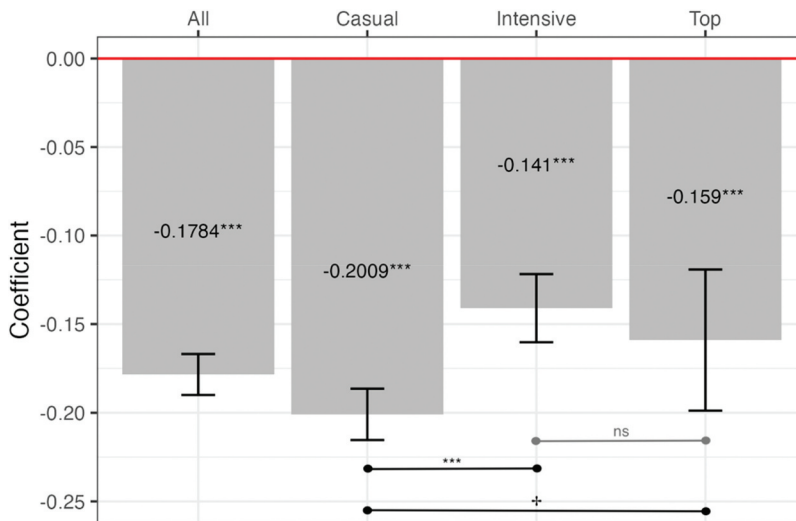


Figure 3. Summary of tests for Hypothesis 1-Hypothesis 1c. Note: ns $p \geq 0.1$. + $p < 0.1$. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$; the bars depict the coefficients of $Treated \times After$ from Table 1. The whiskers show 95% confidence intervals.

where Y_{itwgu} is the readability score of question i asked in tag t in week w by user in group g (treatment or control). We include a set of fixed effects comprising tag σ_t , week-of-the-year μ_w , and tag-week $\mu_w \times \sigma_t$ fixed-effects. We estimate Eq. 3 using a linear model. We utilize our dataset of 3,218,285 questions and their corresponding text to accomplish this. Following the removal of outlier questions (such as those with excessively long/short text or wrong encoding), our dataset contains 3,165,999 questions with text ready to be analyzed. The results can be found in Table 2.

The results displayed in columns (1) to (4) capture the change in complexity (readability score) due to the introduction of ChatGPT. Column (1) presents the overall effect of ChatGPT on complexity. As we can see, the coefficient of $Treated \times After$ is statistically significant and positive, showing that questions become, on average, more complex after the introduction of ChatGPT, thus supporting H2. Next, we test H2a to H2c in a similar fashion as H1a through H1c, namely by interacting $Treated \times After$ with the user group dummy and changing the reference category in each column. Column (2) tests H2a with the reference category of casual users whose effect is reflected in the coefficient of $Treated \times After$. As we see from the positive coefficient of $Treated \times After$, casual users statistically significantly increase the complexity of their questions after the availability of ChatGPT. Importantly, the interaction terms of $Treated \times After$ with *Intensive* and *Top* are the opposite sign (positive) of the effect for casual users and of the same magnitude. This shows that the effect on complexity for intensive and top users is zero; in other words, these groups do not increase their question complexity upon the availability of ChatGPT. This supports H2a.¹⁷

Next, we test H2b and H2c in Columns (3) and (4). In Column (3), we can see again that the effect on complexity is statistically insignificant for intensive users, as shown by the insignificant coefficient of $Treated \times After$. Moreover, top users are not statistically

Table 2. Effect of ChatGPT on question complexity (readability score) across user types on StackExchange.

DV	Question Complexity			
	(1)	(2)	(3)	(4)
Treated	0.00004 (0.00003)	0.0001* (0.00003)	0.00002 (0.00004)	-0.0001 (0.0001)
Treated $\times$ After	0.001*** (0.00004)	0.001*** (0.00005)	0.00001 (0.0001)	0.0001 (0.0001)
Treated $\times$ After $\times$ Casual			0.001*** (0.0001)	0.001*** (0.0001)
Treated $\times$ After $\times$ Intensive		-0.001*** (0.0001)		-0.0001 (0.0001)
Treated $\times$ After $\times$ Top		-0.001*** (0.0001)	0.0001 (0.0001)	
Tag FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Tag $\times$ Week FE	Yes	Yes	Yes	Yes
AskWizard	Yes	Yes	Yes	Yes
User type FE	Yes	Yes	Yes	Yes
Ref. category		Casual	Intensive	Top
Hypothesis test	H2	H2a	H2b	H2c
Observations	3,165,999	3,165,999	3,165,999	3,165,999
R ²	0.104	0.105	0.105	0.105
Adjusted R ²	0.034	0.034	0.034	0.034

Standard errors at the user level in brackets; Q. before (x) by user type: Casual $0 < x < 3$; Intensive $2 < x < 11$; Top $10 > x$. * $p < 0.05$. *** $p < 0.001$. Abbreviation: FE, Fixed Effects; H2, Hypothesis 2.

different from intensive users, as demonstrated by the statistically insignificant interaction term $Treated \times After \times Top$. However, as we have seen before, casual users are statistically significantly different from both groups, increasing the complexity of their questions, thus supporting H2a. We can observe the same pattern in Column (4). Again, the effect on complexity for intensive users is not statistically significantly different from top users, but the impact on casual users supports H2c.¹⁸ Figure 4 summarizes our hypotheses tests visually.

In summary, consistent with our theoretical underpinnings from the CBA and IFT, we find that the questions on SE become more complex after the availability of ChatGPT. This effect is driven by casual users for whom the average question complexity significantly increases, plausibly because they resort to the community for very complex (i.e., high benefit) questions that may not be satisfactorily handled by ChatGPT. As intensive and top users are less likely to engage in cost-benefit evaluation before posting to the community, they do not post more complex questions after the availability of ChatGPT.

Hypotheses 3-3c: Question Novelty

Next, we test whether ChatGPT has affected the novelty of questions on SE, that is, H3-3c.

For this analysis, we incorporate the dependent variable *Question Novelty* on the left side of Eq. 1 and write the following modification:

$$Y_{itw_a u_a w_b u_b g} = \delta_1 Treated_g \times UserType_{u_a} + \delta_2 UserType_{u_a} + \delta_3 Treated_g + \sigma_t + \mu_{w_a} + \eta_{w_b} + \mu_{w_a} \times \sigma_t + \eta_{w_b} \times \sigma_t + \epsilon_{itw_a u_a w_b u_b g} \quad (4)$$

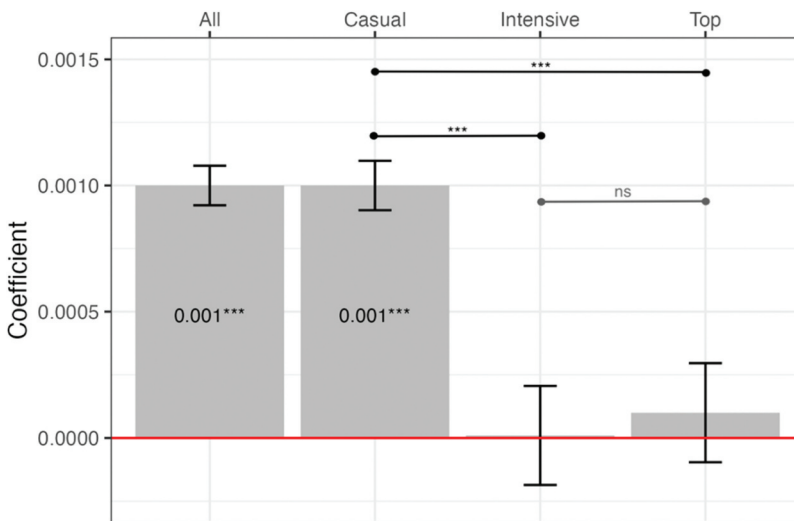


Figure 4. Summary of tests for Hypothesis 2-Hypothesis 2c. Note: ns $p \geq 0.1$. + $p < 0.1$. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$; the bars depict the coefficients of $Treated \times After$ from Table 2. The whiskers show 95% confidence intervals.

where $Y_{itw_a u_a w_b u_b g}$ is the cosine similarity of the pair i of questions in tag asked in the week in the after period w_a by user u_a and the week of the before period w_b by user u_b , of group g (treatment or control). This setting allows us to leverage a rich set of fixed effects comprising tag σ_t , week-of-the-year for the two questions μ_{w_a} and η_{w_b} , and their respective interaction with the tag $\mu_{w_a} \times \sigma_t$ and $\mu_{w_b} \times \sigma_t$.¹⁹

Note that this estimation technique is akin to a DID framework as it estimates the difference in the differences between the vector representations of two questions, that is, their cosine similarity. As in the previous estimations, we interact our focal treatment variable $Treated_g$ with $UserType_{u_a}$ to test for heterogeneous effects. For both the treatment and the control panel, we segment the user type based on the questions asked in the “After” period, that is, on the question asked in a week w_a , by users we denote by u_a . Given that the distribution of question novelty lies in the interval between -1 and 1 , we estimate Eq. 4 using a linear model. Our database has 2,067,233 observations after cleaning it from outliers.

The results of our novelty analysis are displayed in Table 3. The coefficient in Column (1) tests the overall effect, and the focal coefficient is statistically significant and negative. This shows that the questions posted to SE become more novel, on average, after the introduction of ChatGPT.^{20, 21} This provides support for H3. Columns (2) to (4) test H3a to H3c. In Column (2), we see a statistically significant and negative coefficient for the reference category (*Treated*) of casual users, showing that the questions of casual users become more novel after the availability of ChatGPT. By contrast, the interactions with *Intensive* and *Top* are statistically significant and have the opposite sign (positive) compared to casual users. This shows that intensive and top users do not increase the novelty of questions after the availability of ChatGPT, and the questions by intensive users even slightly decrease

Table 3. Effect of ChatGPT on question novelty across user types on StackExchange.

DV	Questions Novelty			
	(1)	(2)	(3)	(4)
Treated	-0.005*** (0.001)	-0.007*** (0.001)	0.001 (0.002)	0.001 (0.003)
Treated × Casual			-0.008*** (0.002)	-0.007* (0.003)
Treated × Intensive		0.008*** (0.002)		0.0005 (0.004)
Treated × Top		0.007* (0.003)	-0.0005 (0.004)	
Tag FE	Yes	Yes	Yes	Yes
Week _{before} FE	Yes	Yes	Yes	Yes
Week _{after} FE	Yes	Yes	Yes	Yes
Tag × Week _{before} FE	Yes	Yes	Yes	Yes
Tag × Week _{after} FE	Yes	Yes	Yes	Yes
Copilot	Yes	Yes	Yes	Yes
User type FE	Yes	Yes	Yes	Yes
Ref. category		Casual	Intensive	Top
Hypothesis test	H3	H3a	H3b	H3c
Observations	2,067,223	2,067,223	2,067,223	2,067,223
R ²	0.179	0.179	0.179	0.179
Adjusted R ²	0.071	0.071	0.071	0.071

Standard errors at the user level in brackets; Q. before (x) by user type: Casual $0 < x < 3$; Intensive $2 < x < 11$; Top $10 > x$. * $p < 0.05$. *** $p < 0.001$. Abbreviation: FE, Fixed Effects; H3, Hypothesis 3.

in novelty ($-0.007 + 0.008 = 0.001$). Thus, it provides support for H3a. Columns (3) and (4) show a very similar pattern. In both Columns, only casual users' questions increase in novelty, whereas the questions of intensive and top users do not. Also, as we can see, the novelty of questions by intensive and top users are not statistically significantly different from each other. This provides support for H3b and H3c. Figure 5 summarizes our hypotheses tests visually.

Some concerns might arise as our novelty indicators might be non-normally distributed, as also noted by Burtch et al. [10]. To address this, we also use a Mann–Whitney U test, a nonparametric test based on the order in which the observations relative to cosine measures might differ between the treatment and the control group. Table 3.2 in Online Supplemental Appendix 3 presents the results and shows a similar pattern to Table 3, namely an increase in novelty in the treatment group relative to the control group.

Discussion and Conclusion

Summary of Results

The release of GenAI had a substantial effect on the usage of the knowledge sharing community SE, which is heterogeneous across casual, intensive, and top users. By analyzing over 3.2 million questions over two years, we find partial to full support for our twelve hypotheses. Table 4 summarizes the results of our hypotheses test. We find that casual users exhibit a stronger decrease in questions than intensive and but not than top users, providing partial support for H1a.²² Interestingly, we find that the decrease in questions by top users is statistically insignificantly different from those by intensive users, thus only partially supporting H1b and rejecting H1c. We surmise this might be

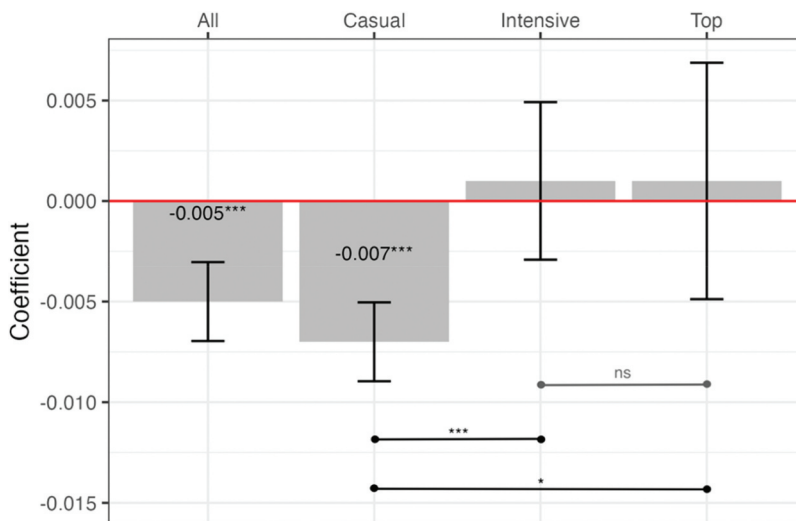


Figure 5. Summary of tests for Hypothesis 3-Hypothesis 3c. Note: ns $p \geq 0.1$. $+p < 0.1$. $*p < 0.05$. $**p < 0.01$. $***p < 0.001$; the bars depict the coefficients of $Treated \times After$ from Table 3. The whiskers show 95% confidence intervals.

Table 4. Summary of hypotheses tests.

Hypothesis	Effect	Result
H1	The availability of Generative AI decrease the number of questions.	Supported
H1a	The decrease in questions by casual users due to the availability of Generative AI is stronger than the decrease by intensive and top users.	Partially Supported
H1b	The decrease in questions by intensive users due to the availability of Generative AI is weaker than the decrease for casual but stronger than the decrease for top users.	Partially Supported
H1c	The decrease in questions by top users due to the availability of Generative AI is weaker than the decrease for casual and intensive users.	Partially Supported
H2	Question complexity increases due to the availability of Generative AI.	Supported
H2a	The increase in question complexity by casual users due to the availability of Generative AI is stronger than the increase by intensive and top users.	Supported
H2b	The increase in question complexity by intensive users due to the availability of Generative AI is weaker than the increase for casual but not different from the increase by top users.	Supported
H2c	The increase in question complexity by top users due to the availability of Generative AI is weaker than the increase for casual but not different from the increase by intensive users.	Supported
H3	Question novelty increases due to the availability of Generative AI.	Supported
H3a	The increase in question novelty by casual users due to the availability of Generative AI is stronger than the increase by intensive and top users.	Supported
H3b	The increase in question novelty by intensive users due to the availability of Generative AI is weaker than the increase for casual but not different from the increase by top users.	Supported
H3c	The increase in question novelty by top users due to the availability of Generative AI is weaker than the increase for casual but not different from the increase by intensive users.	Supported

The conclusion that H1c is partially supported is based on the fact that the robustness check in Online Supplemental Table A1 shows a statistically significant difference at the 1%-level between casual and top users. Abbreviation: AI, artificial intelligence.

because affective and normative commitment across intensive and top users are not mutually exclusive and might overlap. Next, supporting H2 and H3, we find that questions posted to the community after the launch of GenAI exhibit different characteristics compared to those posted before the availability of GenAI. They are more complex in terms of their readability and are more novel. Analyzing the heterogeneity of the effects on complexity and novelty across the user groups shows that the questions of casual users drive the increase in complexity and novelty, supporting H2a-c. In contrast, the questions of intensive and top users neither increase in complexity nor novelty which supports H3a-c.

The combination of these findings allows for some interesting conclusions. First, consistent with our theorizing, the need-based cost-benefit considerations of users driven by continuance commitment decrease the knowledge seeking activity of these users in knowledge sharing communities [5]. Many of them stopped posting questions after the availability of GenAI, which presented a more appealing alternative option in terms of time and effort. Because intensive and top users are, to a higher degree, driven by affective and normative commitment, they are less likely to substitute the community with GenAI and hence exhibit weaker decreases in question.

Second, the average complexity and novelty of questions on SE increased after GenAI was introduced. Our tests of H2a-c and H3a-c trace down increases in question complexity and novelty by casual users as the underlying reason. In contrast, complexity and novelty do not increase for intensive and top users. Consistent with our fusion of the CBA [5] and IFT [48], we reason that this is because, after GenAI, casual users only post questions to the community whose answers promise a high information value, that is, complex and novel

ones. This is because the decreased information collection effort through GenAI makes it even more unattractive for casual users to post questions to the community. We surmise that the questions that casual users post to the community after GenAI are likely questions that GenAI may not satisfactorily answer.

Generalizability

Examining a single platform and a single shock naturally raises questions about the generalizability of the results. These questions, across numerous lines, offer opportunities for future work. First, we examine a knowledge sharing community where users seek viable solutions to the problem described in their questions. The personal gain that users derive is likely a mix stemming from answers and from the commitment to and enjoyment of the community. This mix might be different across communities. On Reddit, for example, users might derive a relatively higher share of personal gain from simply conversing and less from having questions answered. GenAI is arguably not a good substitute for this purpose. Our results are thus more likely to generalize to communities where members exhibit a mix of motivations rather than communities where all members are highly committed and loyal. In other words, the heterogeneous response in questions depends on whether there are heterogeneous commitment types among community members, as per Bateman et al. [5]. Second, GenAI tools such as ChatGPT are capable of providing answers in a very short time. Thus, our results are more likely to generalize to platforms where time pressure is an important factor than to platforms where time is less critical. Finally, and perhaps obviously, a precondition to our results' generalizability is that GenAI can answer the questions at hand; otherwise, we do not expect it to be a good substitute for knowledge-seekers. While the capabilities of GenAI tools are constantly evolving, their performance may worsen in contexts where more domain knowledge is required. For instance, it may be less likely that ChatGPT will affect questions on internal corporate knowledge sharing communities.

Contributions to Research

Our findings make three pivotal contributions to the literature. First, our literature contributes to the nascent stream of research on the effects of GenAI on knowledge sharing communities. While prior research has uncovered that the availability of GenAI reduces knowledge seeking activity in online communities [11, 19], dynamics across heterogeneous user groups have been hitherto overlooked. We thus extend this stream of literature by unveiling heterogeneous dynamics in knowledge seeking activities in response to the availability of GenAI. This is a noteworthy step forward in the literature because, thus far, the prevalent understanding was that GenAI leads to a decline in knowledge seeking activities. While we agree that knowledge seeking activity declines, we add that the decline has noteworthy nuances with respect to which user type reduces knowledge seeking, namely those users with a relatively low commitment to the platform (casual users). In contrast, core users (intensive and top users) reduce their knowledge-seeking activities less drastically. This paints a less bleak picture of the previously assumed effect of GenAI on knowledge seeking activities in online communities. By extension, these results also add to the

nascent literature that has focused on GenAI usage patterns, such as the work characterizing the adopters of GenAI, including their demographics and skills [34, 20]. We contribute to this stream by showing that casual users opt for GenAI instead of posting questions to knowledge sharing communities.

Second, and related to the aforementioned, we further contribute to this stream of research by linking the heterogeneous reduction of knowledge seeking activity to downstream effects on content characteristics. This advances the literature by extending the focus from the question of *how much* toward *what kind* of content, an area where prior literature has been largely silent on [11, 19]. Our findings refine the prevailing narrative that GenAI heralds the demise of knowledge sharing communities by revealing a clear rise in content complexity and novelty. By examining how this transformation unfolds, we uncover a mechanism that offers a counterpoint to prior studies that emphasized the role of central contributors [56, 18, 68, 69, 70]. Instead, our results point to users with looser ties as a driving force behind creating more sophisticated and novel content—broadening our understanding of how knowledge is shaped within online communities.

Third, we contribute to the literature stream on community commitment. Within this well-established stream, a recurring theme is that studies have focused on the ways that commitment is formed [54, 55] and how commitment can be a lever to nurture community longevity, for example, in terms of content growth [5, 63] or willingness-to-pay for additional community features [46]. Notably, these studies frame commitment as an engine to grow communities. By contrast, we contribute an additional way to view commitment, namely as a protective shield against competitors from beyond the boundaries of the community, in our case, GenAI. This is important because it presents the literature on community commitment in a new light, namely one in which the commitment type, such as affective or normative, may provide a competitive edge against GenAI alternatives. In a nutshell, we contribute a new perspective that is key to protecting communities from GenAI rivals by studying the literature on community commitment.

Contributions to Practice

First, while it is self-evident that the reduction of knowledge seeking due to GenAI presents a threat to the survivability of platforms, our results also offer a glimpse of hope. Our estimates suggest a less pronounced reduction for advanced and top users. This implies that the platform might be able to at least partially retain its dedicated core user base. On the one hand, this is important because studies have highlighted the critical role of core users on platforms [56, 18]. Our finding that questions become more complex and novel, arguably a signal for improved content quality, is another positive implication for platforms. On the other hand, the rise of GenAI might marginalize casual or new users on the platform. This presents opportunities as well as challenges for the platform. For one, platforms need a continuous inflow of new users who adopt the platform over the long term to remain attractive. Second, the platform may try to monetize the improved content quality. SE's recent collaboration with OpenAI might be one step in this direction [59].²³ Yet, our results suggest this collaboration may be a double-edged sword. Recent user discussions have hotly debated this collaboration [52]. Users state that this would further demotivate content contribution, knowing their voluntary content is used for company profits. At the same time, ChatGPT might become even more potent with this data, potentially leading to an

even stronger substitution of knowledge sharing communities by ChatGPT. The measure to provide attribution and referrals to SE content within ChatGPT answers might be partially effective in rerouting traffic from ChatGPT to SE and meaningfully acknowledging contributors.

Second, our results provide several implications for GenAI developers. First, the official collaboration between SE and OpenAI [59] indicates that SE is an important source of training data for ChatGPT. Early results from the GenAI training literature indicate that the performance of GenAIs may decrease over time [15] if high-quality training data is not added. Against this backdrop, and if we accept the assumption that SE continuously serves as ChatGPT training data, the launch of ChatGPT has arguably increased its own training data quality as questions become more complex and novel. Consequently, the interplay between SE and ChatGPT might prove to be a virtuous cycle for OpenAI: Its launch has improved the training data, which it can use to improve its product. Nevertheless, our finding on the decrease in the number of questions also calls for caution in exploiting this cycle. As ChatGPT further improves, data on SE might decrease even more, and with increasing ChatGPT performance, even the high-quality content might increasingly disappear. Consequently, managing the relationship between SE and OpenAI requires a delicate balancing act. This is further exacerbated by users' discontent about their voluntary contributions being used by OpenAI [52]. Hence, to mitigate declines in contributions, GenAI developers should find a way to acknowledge and reward voluntary knowledge contributions.

Third, and related to the aforementioned, our results also carry weight for community members and society as a whole. To the extent that community members are data workers that provide public goods [41, 40], we cautiously surmise that our results of increased content complexity and novelty may potentially benefit the negotiating power of community members. From a user perspective, the change in content—less volume but higher complexity and novelty of content—might also present a bright side to the usage experience of the community members. Particularly, users may find it easier to sift through the platform to find the sophisticated questions and their answers because fewer but more novel questions are flowing in after the availability of GenAI.

Limitations and Future Research

Naturally, our work is not without limitations. First, there are limits on how closely we can identify the theoretical mechanisms underlying our hypotheses' development because we use observational data. For example, we cannot directly observe the commitment types from the CBA [5] of the users in our sample. Consistent with prior research [54, 5], follow-up research might leverage survey instruments to examine this further. A more precise dissection of the commitment types might explain the insignificant difference between intensive and top users.

Second, we categorize users based on their question asking frequency, which might exclude or underrepresent users who are highly committed to the community but not necessarily frequent question askers. Therefore, this categorization might overlook nuances, for instance, within the top users, particularly those who still engage with the platform in activities beyond merely asking questions. Hence, future research might use a broader variety of activities to segment user groups to further complement our results.

Third, there are natural limitations to the time frame of data we can study, especially with our DiD design, which requires equal additional proportions of pre-treatment observations for each additional post-treatment observation. As time goes by, studies could use a longer time frame of data, potentially with a different causal identification design.

Finally, while we use the launch of ChatGPT as the treatment in our estimation framework, other GenAIs are available to users (e.g., Bing AI, Bard, and more). Our effects might also be partially driven by them and not solely by ChatGPT. Therefore, our estimates represent an upper bound of the effects of ChatGPT.

Conclusions

This study offers insights into how GenAI tools, such as ChatGPT, reshape the knowledge seeking behavior and content formation in Question and Answer (Q&A) communities. Drawing on the commitment-based approach and information foraging theory, we show that the drop in community engagement is neither uniform nor definitively catastrophic: casual, cost-benefit-driven users are more prone to reduce their activity, whereas members with higher commitment exhibit a smaller decline. Simultaneously, the remaining content becomes more complex and novel, suggesting that GenAI's impact can enhance specific dimensions of knowledge creation. These findings stress the importance of accounting for heterogeneous user motivations when assessing the broader implications of GenAI on digital communities. More generally, our study underlines the need for ongoing research into the evolving interplay between GenAI and community-based knowledge sharing.

Notes

1. We acknowledge that the ladder of participation recognizes four different user categories, but since one of them is the category of lurkers who do not contribute content, we do not include this category as their behavior is not observable.
2. This literature posits that communities consist of 90% low-activity users, 9% medium-activity users, and 1% high-activity users. We thank an anonymous reviewer for pointing us toward this literature.
3. As we will explain later when we define our variables, we will differentiate casual, intensive, and top users based on their number of questions posted to the community before the availability of GenAI.
4. Although the average effect is not the focus of our paper since it was already investigated in previous work [11, 19], we retain this hypothesis as a point of departure.
5. We exclude four weeks of data (week 21 to week 24) to prevent potential contamination between our control and treatment period (for additional explanation, see the General Identification Strategy section).
6. We discard the subforum dedicated to Mathematical questions as it split into two different subforums during the observation period.
7. Reassuringly, this operationalization of user types is also consistent when inspecting the number of answers of these groups. The average number of answers is 0.6 for casual users, 2.1 for intensive users, and 5.9 for top users.
8. We thank the Associate Editor for this pointer.
9. See https://www.sbert.net/docs/pretrained_models.html and <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2> for more information
10. Each question on SE has at least one tag, which is like a topic classification of a question, such as *csv*, *python*, or *ubuntu*, for instance. For computational feasibility, we focus on tags with

observations in every week of our database (421 tags, corresponding to approximately 85% of the observations).

11. The variable $After_w$ is completely absorbed by the week fixed effects.
12. Note that users can be in both the treated (2022-23) and the control (2021-22) group.
13. We compute the Pearson dispersion statistics [43] and find limited evidence of overdispersion ($P_{disp} = 1.4$). However, we also implement this equation for additional robustness using a negative binomial model. The results are consistent and displayed in Table A.1 in Online Supplemental Appendix 1.
14. In Online Supplemental Appendix 1, we provide a formal test of the parallel trends assumption in Figure A.1. We also repeat this test for each user type in Figures A.2 for casual users, A.3 for intensive users, and A.4 for top users.
15. We compute a percentage point increase d such that $100(e^a - 1) + d = 100(e^{a+b} - 1)$, which gives us $d = 100(e^{a+b} - e^a)$.
16. We provide additional analyses in the Online Supplemental Appendix. In Online Supplemental Appendix 4, we examine whether casual users stop asking questions altogether (extensive margin, following [31]) or ask fewer questions (intensive margin). We find that casual users exhibit a 24% decline (Table D.1) and drive the extensive margin change, while the overall and homogeneous intensive-margin reduction is 1.5% (Table D.2). We also investigate whether dropping out of asking questions affects answering activity, finding a homogeneous 5% decrease (Table D.3 of Online Supplemental Appendix 4).
17. Interestingly, we note that even before the availability of ChatGPT, casual users were asking, on average, the most complex questions of the three groups. Their questions' mean complexity is 0.0281, compared to 0.0275 (intensive users) and 0.0269 (top users). This supports our theorizing that even in the absence of ChatGPT, casual users only post questions to the community whose answers have a high information value, given their cost-benefit considerations.
18. In Online Supplemental Appendix 2, we provide a formal test of the parallel trends assumption in Figure B.1. For further robustness, we also present the estimation results of the average effect separately for the different readability metrics in Table B.1 in Online Supplemental Appendix 2. Finally, we control for the effect of ChatGPT usage and bans in Figure E.1 of Online Supplemental Appendix 5.
19. One difference to our previous models, though, is that the presence of AskWizard is controlled for via the week fixed-effects in the specification.
20. Recall that the dependent variable is cosine similarity. Therefore, a negative sign means decreasing similarity and, thus, higher novelty.
21. For additional robustness, we also present the results for three alternative methods of operationalizing the cosine similarity in Table C.1 in Online Supplemental Appendix 3.
22. While the difference between casual and top users is not significant at the 5%, it is significant at the 1%-level in the robustness check in Table 1.1 in the Online Supplemental Appendix 1 that uses a negative binomial model. Moreover, the magnitude of this difference is quite substantial with 3.5%, which is approximately one-fifth of the effect on casual users.
23. On May 6, 2024, SE and OpenAI announced their collaboration, which granted OpenAI exclusive access to SE content via an API, whereas SE would be able to harness OpenAI products and benefit from content attribution and referrals to SE from ChatGPT.

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